Combined Control Strategy for an Inverted Pendulum on a Cart

1 Introduction

This document details a hybrid control approach that combines three control strategies to stabilize an inverted pendulum on a cart:

1. **LQR Control** for baseline stabilization.
2. **Backstepping Control** to provide additional nonlinear compensation.
3. **Sliding Mode Control (SMC)** to robustly reject uncertainties and disturbances.

A low-pass filter is applied to the combined control signal to reduce high-frequency chattering and ensure smooth actuator commands. Each component has its own set of tuning parameters that influence overall system performance.

2 System Model Overview

The plant is represented by the state-space model:

$$A = \begin{bmatrix} 0 & 1 & 0 & 0 \\ 0 & 0 & -\frac{b\,d}{\Delta} & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & \frac{a\,d}{\Delta} & 0 \end{bmatrix}, \quad B = \begin{bmatrix} 0 \\ \frac{c}{\Delta} \\ 0 \\ -\frac{b}{\Delta} \end{bmatrix},$$

$$C = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 \end{bmatrix}, \quad D = \begin{bmatrix} 0 \\ 0 \end{bmatrix},$$

with the common parameters defined as:

$$a = 2m_w + m_p + \frac{2I_w}{r^2}, \quad b = m_p\,l, \quad c = m_p\,l^2 + I_p,$$
$$d = m_p\,g\,l, \quad \Delta = a\,c - b^2.$$

The numerical values for the physical parameters are:

$$m_w = 0.432 \text{ kg}, \quad m_p = 5.0 \text{ kg}, \quad l = 0.4 \text{ m}, \quad r = 0.0726 \text{ m},$$
$$g = 9.81 \text{ m/s}^2, \quad I_w = 0.5 \times 0.432 \times (0.0726)^2, \quad I_p = 5 \times (0.4)^2.$$

3 Controller Components and Tuning Parameters

3.1 LQR Control

Purpose: Provides a baseline stabilizing control signal based on a quadratic cost minimization.

Design:

$$Q = \text{diag}(100,\ 1,\ 1000,\ 1), \quad R = 0.01.$$

The LQR gain K is computed using the standard LQR synthesis:

$$u_{\text{lqr}} = -K\,x.$$

Tuning:

- Increasing the weight on the cart position (first element of Q) forces a tighter regulation of x_1.
- Increasing the weight on the pendulum angle (third element of Q) improves the stabilization of x_3 but may result in higher control effort.
- R penalizes the magnitude of the control input; a smaller R yields a more aggressive controller.

3.2 Sliding Mode Control (SMC)

Purpose: Enhances robustness by providing fast corrective action to reject uncertainties and disturbances.

Design: A sliding surface is defined as:

$$s = x_3 + \lambda\,x_4,$$

where:

- λ is the sliding surface slope. A lower λ reduces sensitivity to angular velocity.

The SMC law is implemented with a smoothing factor using the hyperbolic tangent:

$$u_{\text{smc}} = -\eta\,\tanh\left(\frac{s}{\epsilon}\right).$$

Tuning:

- λ: Determines the relative weighting between the pendulum angle and angular velocity. A properly chosen λ balances the control action between reducing angle error and damping.
- η: The reaching gain. Increasing η forces the state to reach the sliding surface faster but may induce higher control activity.
- ϵ: The smoothness factor in the tanh function. A smaller ϵ makes the control law closer to a discontinuous sign function (potentially causing chattering), while a larger ϵ smooths the control input.

3.3 Backstepping Control

Purpose: Provides additional nonlinear compensation by designing virtual control laws based on the system states.

Design: Backstepping control is implemented via the following virtual control laws:

$$\begin{aligned} v_1 &= -k_1\, x_3, \\ v_2 &= -k_2\, x_4, \\ v_3 &= -k_3\, (x_1 + x_2), \\ v_4 &= -k_4\, (x_3 + x_4). \end{aligned}$$

The combined backstepping control is:

$$u_{\mathrm{bs}} = v_1 + v_2 + v_3 + v_4.$$

Tuning:

- k_1, k_2, k_3, k_4: These gains are tuned to ensure that each virtual control law contributes to reducing the error in its corresponding state variable.
- Higher gains will generally result in faster error convergence but may lead to increased control effort and potential overshoot.

3.4 Combined Control Strategy

The overall control signal is a weighted sum of the three components:

$$u_{\mathrm{raw}} = \alpha\, u_{\mathrm{lqr}} + \beta\, u_{\mathrm{bs}} + \gamma\, u_{\mathrm{smc}},$$

where:

- α adjusts the influence of the LQR component.
- β weights the contribution of the backstepping controller.
- γ scales the effect of the sliding mode controller.

Tuning:

- Increasing α prioritizes the baseline performance of the LQR, ensuring overall system stability.
- Increasing β emphasizes nonlinear compensation from the backstepping layer.
- Increasing γ strengthens the SMC action for robust disturbance rejection.

3.5 Low-Pass Filtering

Purpose: To smooth the combined control signal and reduce high-frequency noise and chattering.

Design: A first-order low-pass filter is applied:

$$\text{filtered\_u} = (1 - \alpha_{filter})\, \text{filtered\_u} + \alpha_{filter}\, u_{\text{raw}},$$

where α_{filter} is the filter coefficient.

Tuning:

- A smaller α_{filter} results in slower response (more smoothing) but can delay control actions.
- A larger α_{filter} allows the control signal to track rapid changes but may let through more high-frequency noise.

4 Simulation and Implementation

Simulation Setup:

- The simulation uses a fixed time step $t_s = 0.001$ seconds over a total time of 5 seconds.
- The initial state is chosen as:

  $$x(0) = [0; 0; 0.1; 0],$$

  indicating a small initial pendulum angle deviation.

State Update: The closed-loop system dynamics are integrated using Euler integration:

$$x_{k+1} = x_k + t_s \Big(A\, x_k + B\, u_k \Big),$$

where u_k is the filtered control input applied at each time step.

5 Summary

The combined control strategy leverages the strengths of three distinct controllers:

a. **LQR:** Provides a solid baseline by minimizing a quadratic cost, with performance tuned via Q and R.

b. **Backstepping:** Offers additional nonlinear compensation using gains k_1, k_2, k_3, and k_4, which are tuned to accelerate error convergence.

c. **Sliding Mode Control:** Enhances robustness to disturbances and uncertainties using parameters λ, η, and ϵ to drive the system towards a designed sliding surface.

The overall control input is formed as a weighted sum:

$$u_{\mathrm{raw}} = \alpha\, u_{\mathrm{lqr}} + \beta\, u_{\mathrm{bs}} + \gamma\, u_{\mathrm{smc}},$$

which is then filtered to produce a smooth control effort. Tuning of the weights α, β, γ, and the filter coefficient α_{filter} is crucial to achieve the desired balance between responsiveness, robustness, and smooth actuation.